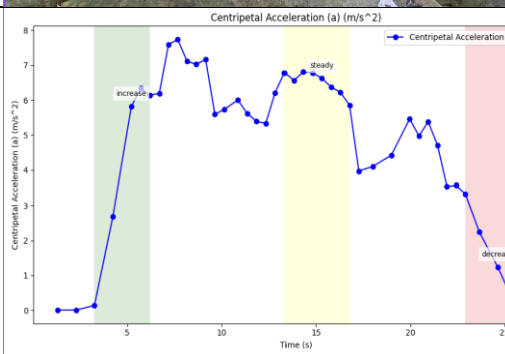
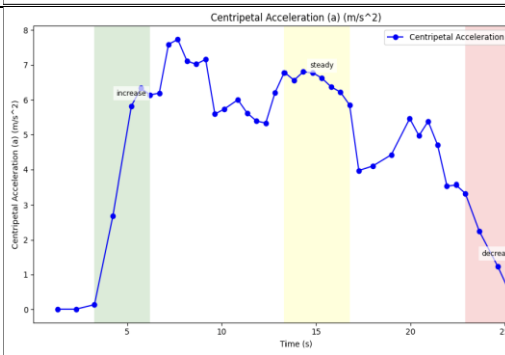
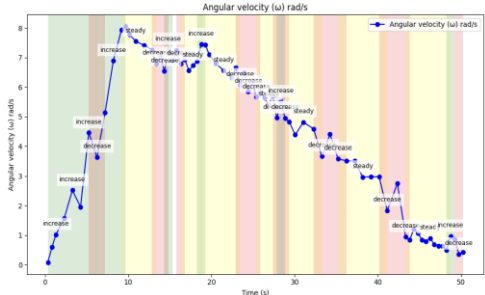
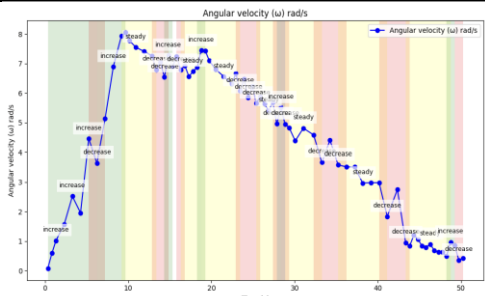
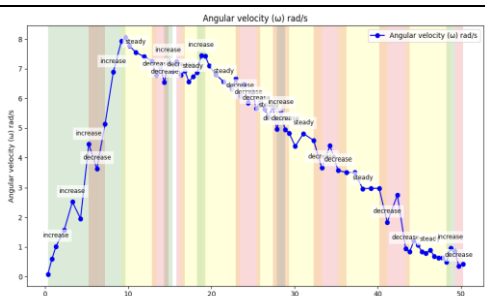


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1		Sebuah waist trainer berputar dengan kecepatan sudut tetap (ω) dalam kondisi stabil. Jari-jarinya (r) adalah 0,09 m, dan antara detik ke-7 hingga ke-9, percepatan sentripetal (a) teramati tetap sekitar 15 m/s². Mengingat periode yang stabil ini, bagaimana kecepatan sudut waist trainer (ω) memengaruhi percepatan sentripetal (a) yang diamati? Bisakah kamu menjelaskan hubungan antara variabel-variabel ini dalam selang waktu tersebut?																																																																																																																																																																								
2		Dalam rentang waktu antara detik ke-7 hingga ke-10, percepatan sentripetal (a) dari waist trainer teramati stabil sekitar 15 m/s². Apa yang bisa dikatakan tentang perilaku kecepatan sudut (ω) selama periode ini?																																																																																																																																																																								
3		Seorang siswa mengamati gerakan waist trainer. Waist trainer tersebut bergerak dalam lintasan melingkar dengan jari-jari (r) sebesar 0,09 m. Jika siswa tersebut mengukur kecepatan sudut (ω) dari gerakan waist trainer, dapatkah kamu jelaskan arah dari kecepatan sudut ?																																																																																																																																																																								
4		Jika Sarah sedang mempelajari mekanisme waist trainer, dan ia mengukur jari-jari (r) dari pusat kipas ke tepi waist trainer sebesar 0,09 m. Ia mengamati bahwa alat tersebut menyelesaikan satu putaran penuh dalam 0,5 detik. Bisakah kamu membantu Sarah menunjukkan rumus hubungan antara variabel-variabel tersebut?																																																																																																																																																																								
5	<table><tr><th>Time (s)</th><th>Acceleration (m/s²)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>0.35</td><td>0.03</td><td>0.01</td><td>Steady</td></tr><tr><td>0.84</td><td>0.01</td><td>0</td><td>Steady</td></tr><tr><td>1.33</td><td>0</td><td>0</td><td>Steady</td></tr><tr><td>2.31</td><td>0.23</td><td>2.74</td><td>Increase</td></tr><tr><td>3.3</td><td>6.15</td><td>8.85</td><td>Increase</td></tr><tr><td>4.28</td><td>13.34</td><td>11.6</td><td>Increase</td></tr><tr><td>5.26</td><td>13.65</td><td>12.71</td><td>Increase</td></tr><tr><td>6.25</td><td>14.32</td><td>12.7</td><td>Steady</td></tr><tr><td>7.23</td><td>15.04</td><td>13.33</td><td>Steady</td></tr><tr><td>8.21</td><td>15.79</td><td>14.24</td><td>Steady</td></tr><tr><td>9.2</td><td>14.37</td><td>14.65</td><td>Steady</td></tr><tr><td>10.18</td><td>16.36</td><td>14.17</td><td></td></tr><tr><td>11.16</td><td>14.87</td><td>14</td><td></td></tr><tr><td>12.14</td><td>14.33</td><td>13.27</td><td></td></tr><tr><td>12.88</td><td>14.53</td><td>14.79</td><td></td></tr><tr><td>13.37</td><td>15.81</td><td>13.86</td><td></td></tr><tr><td>13.86</td><td>14.31</td><td>13.76</td><td></td></tr><tr><td>14.35</td><td>15.14</td><td>13.75</td><td>Steady</td></tr><tr><td>14.84</td><td>15.01</td><td>13.18</td><td>Steady</td></tr><tr><td>15.34</td><td>15.17</td><td>13.76</td><td>Steady</td></tr><tr><td>15.83</td><td>14.52</td><td>12.41</td><td>Steady</td></tr><tr><td>16.32</td><td>14.15</td><td>12.85</td><td>Steady</td></tr><tr><td>16.81</td><td>14.38</td><td>12.94</td><td>Steady</td></tr><tr><td>17.3</td><td>14.33</td><td>12.4</td><td>Steady</td></tr><tr><td>17.79</td><td>13.59</td><td>11.87</td><td></td></tr><tr><td>18.28</td><td>12.93</td><td>11.21</td><td></td></tr><tr><td>18.78</td><td>12.16</td><td>10.66</td><td>Steady</td></tr><tr><td>19.27</td><td>11.72</td><td>10.22</td><td>Steady</td></tr><tr><td>19.76</td><td>11.28</td><td>9.77</td><td>Steady</td></tr><tr><td>20.25</td><td>10.68</td><td>9.19</td><td>Steady</td></tr><tr><td>20.74</td><td>9.55</td><td>8.05</td><td>Steady</td></tr><tr><td>21.23</td><td>8.89</td><td>7.64</td><td>Steady</td></tr><tr><td>21.73</td><td>8.16</td><td>7.05</td><td>Steady</td></tr><tr><td>22.22</td><td>7.48</td><td>6.58</td><td>Steady</td></tr><tr><td>22.95</td><td>6.71</td><td>5.31</td><td></td></tr><tr><td>23.44</td><td>5.98</td><td>3.84</td><td>Decrease</td></tr><tr><td>23.93</td><td>5.07</td><td>2.83</td><td>Decrease</td></tr><tr><td>24.42</td><td>4.43</td><td>2.52</td><td>Decrease</td></tr><tr><td>24.92</td><td>2.36</td><td>2.15</td><td>Decrease</td></tr><tr><td>25.41</td><td>1.73</td><td>0.93</td><td>Decrease</td></tr><tr><td>25.9</td><td>1.52</td><td>0.01</td><td></td></tr></table>	Time (s)	Acceleration (m/s ²)	Angular Velocity (rad/s)	Phase	0.35	0.03	0.01	Steady	0.84	0.01	0	Steady	1.33	0	0	Steady	2.31	0.23	2.74	Increase	3.3	6.15	8.85	Increase	4.28	13.34	11.6	Increase	5.26	13.65	12.71	Increase	6.25	14.32	12.7	Steady	7.23	15.04	13.33	Steady	8.21	15.79	14.24	Steady	9.2	14.37	14.65	Steady	10.18	16.36	14.17		11.16	14.87	14		12.14	14.33	13.27		12.88	14.53	14.79		13.37	15.81	13.86		13.86	14.31	13.76		14.35	15.14	13.75	Steady	14.84	15.01	13.18	Steady	15.34	15.17	13.76	Steady	15.83	14.52	12.41	Steady	16.32	14.15	12.85	Steady	16.81	14.38	12.94	Steady	17.3	14.33	12.4	Steady	17.79	13.59	11.87		18.28	12.93	11.21		18.78	12.16	10.66	Steady	19.27	11.72	10.22	Steady	19.76	11.28	9.77	Steady	20.25	10.68	9.19	Steady	20.74	9.55	8.05	Steady	21.23	8.89	7.64	Steady	21.73	8.16	7.05	Steady	22.22	7.48	6.58	Steady	22.95	6.71	5.31		23.44	5.98	3.84	Decrease	23.93	5.07	2.83	Decrease	24.42	4.43	2.52	Decrease	24.92	2.36	2.15	Decrease	25.41	1.73	0.93	Decrease	25.9	1.52	0.01		Berdasarkan pola yang diamati dalam eksperimen waist trainer, di mana kecepatan sudut (ω) dan percepatan sentripetal (a) meningkat dan menurun bersama-sama, bagaimana hal ini mencerminkan prinsip bahwa percepatan sentripetal berbanding lurus dengan kuadrat dari kecepatan sudut?
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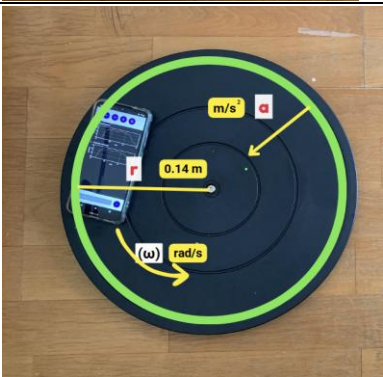
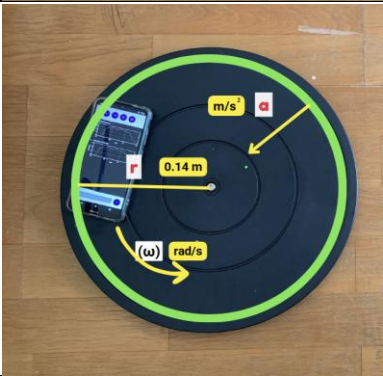
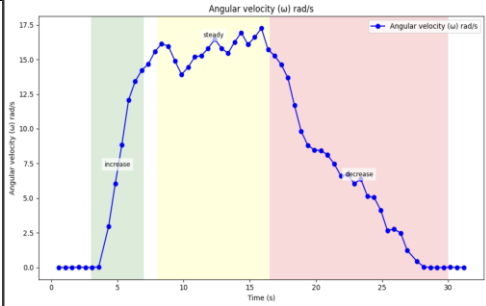
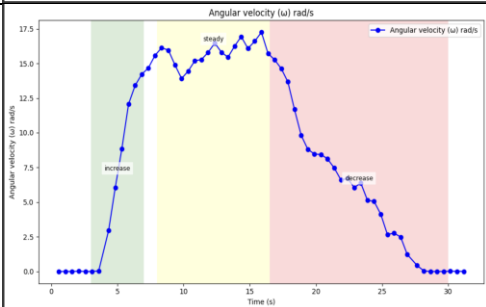
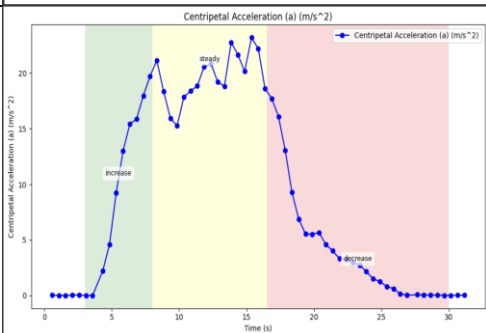
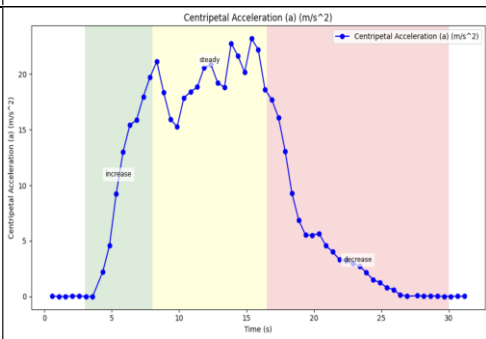
7		Sebuah roda sedang berputar dengan kecepatan tetap. Jika seorang siswa meningkatkan kecepatan sudut (ω) dari roda tersebut, apa dampaknya terhadap percepatan sentripetal (a)? Apakah percepatan sentripetal akan meningkat, menurun, atau tetap sama? Jari-jari roda adalah 0,25 m.																																																																																																																																																																												
8		Kamu sedang mengamati sebuah roda dengan jari-jari 0,25 m. Saat roda berputar pada kecepatan tertingginya, ujung roda bergerak dengan kecepatan sudut (ω). Dapatkah kamu menentukan percepatan sentripetal (a) dari ujung roda tersebut? Harap nyatakan jawabanmu dalam bentuk ω .																																																																																																																																																																												
9		Selama periode stabil antara detik ke-13 hingga ke-17 dalam grafik, jika diketahui jari-jari roda adalah 0,25 m, dapatkah kamu memperkirakan apakah kecepatan sudut (ω) roda tersebut tinggi atau rendah berdasarkan percepatan sentripetal (a) yang relatif stabil sekitar 7 m/s ² ?																																																																																																																																																																												
10		Selama periode stabil dari detik ke-13 hingga ke-17, percepatan sentripetal (a) dari kipas listrik teramati tetap sekitar 7 m/s ² . Apa yang bisa disimpulkan mengenai kecepatan sudut (ω) dari roda tersebut selama periode ini?																																																																																																																																																																												
11	<table><tr><th>Time (s)</th><th>Acceleration (m/s²)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>1.29</td><td>0.01</td><td>0</td><td></td></tr><tr><td>2.28</td><td>0.01</td><td>0</td><td></td></tr><tr><td>3.26</td><td>0.14</td><td>0.23</td><td>Increase</td></tr><tr><td>4.24</td><td>2.69</td><td>3.7</td><td>Increase</td></tr><tr><td>5.22</td><td>5.82</td><td>4.08</td><td>Increase</td></tr><tr><td>5.72</td><td>6.33</td><td>5.32</td><td>Increase</td></tr><tr><td>6.21</td><td>6.14</td><td>5.32</td><td>Increase</td></tr><tr><td>6.7</td><td>6.19</td><td>4.52</td><td></td></tr><tr><td>7.19</td><td>7.59</td><td>5.97</td><td></td></tr><tr><td>7.68</td><td>7.73</td><td>5.96</td><td></td></tr><tr><td>8.17</td><td>7.11</td><td>6.52</td><td></td></tr><tr><td>8.66</td><td>7.02</td><td>5.53</td><td></td></tr><tr><td>9.16</td><td>7.16</td><td>5.58</td><td></td></tr><tr><td>9.65</td><td>5.59</td><td>5.13</td><td></td></tr><tr><td>10.14</td><td>5.73</td><td>4.54</td><td></td></tr><tr><td>10.87</td><td>6</td><td>4.26</td><td></td></tr><tr><td>11.36</td><td>5.62</td><td>4.32</td><td></td></tr><tr><td>11.86</td><td>5.39</td><td>5.88</td><td></td></tr><tr><td>12.35</td><td>5.33</td><td>5.73</td><td></td></tr><tr><td>12.84</td><td>6.2</td><td>5.67</td><td></td></tr><tr><td>13.33</td><td>6.78</td><td>5.85</td><td>Steady</td></tr><tr><td>13.82</td><td>6.56</td><td>6.48</td><td>Steady</td></tr><tr><td>14.31</td><td>6.8</td><td>6.86</td><td>Steady</td></tr><tr><td>14.8</td><td>6.78</td><td>6.45</td><td>Steady</td></tr><tr><td>15.3</td><td>6.62</td><td>6.94</td><td>Steady</td></tr><tr><td>15.79</td><td>6.37</td><td>6.37</td><td>Steady</td></tr><tr><td>16.28</td><td>6.22</td><td>6.59</td><td>Steady</td></tr><tr><td>16.77</td><td>5.85</td><td>6.15</td><td>Steady</td></tr><tr><td>17.26</td><td>3.97</td><td>6.03</td><td>Steady</td></tr><tr><td>18</td><td>4.1</td><td>6.02</td><td></td></tr><tr><td>18.98</td><td>4.42</td><td>4.02</td><td></td></tr><tr><td>19.96</td><td>5.46</td><td>3.89</td><td></td></tr><tr><td>20.45</td><td>4.97</td><td>1.21</td><td></td></tr><tr><td>20.94</td><td>5.38</td><td>1.57</td><td></td></tr><tr><td>21.44</td><td>4.7</td><td>1.05</td><td></td></tr><tr><td>21.93</td><td>3.54</td><td>1.04</td><td></td></tr><tr><td>22.42</td><td>3.57</td><td>1.28</td><td></td></tr><tr><td>22.91</td><td>3.33</td><td>1.12</td><td></td></tr><tr><td>23.65</td><td>2.25</td><td>0</td><td>Decrease</td></tr><tr><td>24.63</td><td>1.24</td><td>0.15</td><td>Decrease</td></tr><tr><td>25.62</td><td>0.08</td><td>0.4</td><td>Decrease</td></tr><tr><td>26.6</td><td>0.05</td><td>0.37</td><td>Decrease</td></tr></table>	Time (s)	Acceleration (m/s ²)	Angular Velocity (rad/s)	Phase	1.29	0.01	0		2.28	0.01	0		3.26	0.14	0.23	Increase	4.24	2.69	3.7	Increase	5.22	5.82	4.08	Increase	5.72	6.33	5.32	Increase	6.21	6.14	5.32	Increase	6.7	6.19	4.52		7.19	7.59	5.97		7.68	7.73	5.96		8.17	7.11	6.52		8.66	7.02	5.53		9.16	7.16	5.58		9.65	5.59	5.13		10.14	5.73	4.54		10.87	6	4.26		11.36	5.62	4.32		11.86	5.39	5.88		12.35	5.33	5.73		12.84	6.2	5.67		13.33	6.78	5.85	Steady	13.82	6.56	6.48	Steady	14.31	6.8	6.86	Steady	14.8	6.78	6.45	Steady	15.3	6.62	6.94	Steady	15.79	6.37	6.37	Steady	16.28	6.22	6.59	Steady	16.77	5.85	6.15	Steady	17.26	3.97	6.03	Steady	18	4.1	6.02		18.98	4.42	4.02		19.96	5.46	3.89		20.45	4.97	1.21		20.94	5.38	1.57		21.44	4.7	1.05		21.93	3.54	1.04		22.42	3.57	1.28		22.91	3.33	1.12		23.65	2.25	0	Decrease	24.63	1.24	0.15	Decrease	25.62	0.08	0.4	Decrease	26.6	0.05	0.37	Decrease	Berdasarkan analisis data yang diberikan, dapatkah kamu menjelaskan mengapa mungkin terjadi fluktuasi dalam korelasi antara percepatan sentripetal (a) dan kecepatan sudut (ω) dari roda, terutama selama rentang waktu 5,22 s hingga 6,7 s?
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13	<table><tr><th>Time (s)</th><th>Acceleration (m/s^2)</th><th>Angular Velocity (rad/s)</th></tr><tr><td>1.29</td><td>0.01</td><td>0</td></tr><tr><td>2.28</td><td>0.01</td><td>0</td></tr><tr><td>3.26</td><td>0.14</td><td>0.23</td></tr><tr><td>4.24</td><td>2.69</td><td>3.7</td></tr><tr><td>5.22</td><td>5.82</td><td>4.08</td></tr><tr><td>5.72</td><td>6.33</td><td>5.32</td></tr><tr><td>6.21</td><td>6.14</td><td>5.32</td></tr><tr><td>6.7</td><td>6.19</td><td>4.52</td></tr><tr><td>7.19</td><td>7.59</td><td>5.97</td></tr><tr><td>7.68</td><td>7.73</td><td>5.96</td></tr><tr><td>8.17</td><td>7.11</td><td>6.52</td></tr><tr><td>8.66</td><td>7.02</td><td>5.53</td></tr><tr><td>9.16</td><td>7.16</td><td>5.58</td></tr><tr><td>9.65</td><td>5.59</td><td>5.13</td></tr><tr><td>10.14</td><td>5.73</td><td>4.54</td></tr><tr><td>10.87</td><td>6</td><td>4.26</td></tr><tr><td>11.36</td><td>5.62</td><td>4.32</td></tr><tr><td>11.86</td><td>5.39</td><td>5.88</td></tr><tr><td>12.35</td><td>5.33</td><td>5.73</td></tr><tr><td>12.84</td><td>6.2</td><td>5.67</td></tr><tr><td>13.33</td><td>6.78</td><td>5.85</td></tr><tr><td>13.82</td><td>6.56</td><td>6.48</td></tr><tr><td>14.31</td><td>6.8</td><td>6.86</td></tr><tr><td>14.8</td><td>6.78</td><td>6.45</td></tr><tr><td>15.3</td><td>6.62</td><td>6.94</td></tr><tr><td>15.79</td><td>6.37</td><td>6.37</td></tr><tr><td>16.28</td><td>6.22</td><td>6.59</td></tr><tr><td>16.77</td><td>5.85</td><td>6.15</td></tr><tr><td>17.26</td><td>3.97</td><td>6.03</td></tr><tr><td>18</td><td>4.1</td><td>6.02</td></tr><tr><td>18.98</td><td>4.42</td><td>4.02</td></tr><tr><td>19.96</td><td>5.46</td><td>3.89</td></tr><tr><td>20.45</td><td>4.97</td><td>1.21</td></tr><tr><td>20.94</td><td>5.38</td><td>1.57</td></tr><tr><td>21.44</td><td>4.7</td><td>1.05</td></tr><tr><td>21.93</td><td>3.54</td><td>1.04</td></tr><tr><td>22.42</td><td>3.57</td><td>1.28</td></tr><tr><td>22.91</td><td>3.33</td><td>1.12</td></tr><tr><td>23.65</td><td>2.25</td><td>0</td></tr><tr><td>24.63</td><td>1.24</td><td>0.15</td></tr><tr><td>25.62</td><td>0.08</td><td>0.4</td></tr><tr><td>26.6</td><td>0.05</td><td>0.37</td></tr></table>	Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	1.29	0.01	0	2.28	0.01	0	3.26	0.14	0.23	4.24	2.69	3.7	5.22	5.82	4.08	5.72	6.33	5.32	6.21	6.14	5.32	6.7	6.19	4.52	7.19	7.59	5.97	7.68	7.73	5.96	8.17	7.11	6.52	8.66	7.02	5.53	9.16	7.16	5.58	9.65	5.59	5.13	10.14	5.73	4.54	10.87	6	4.26	11.36	5.62	4.32	11.86	5.39	5.88	12.35	5.33	5.73	12.84	6.2	5.67	13.33	6.78	5.85	13.82	6.56	6.48	14.31	6.8	6.86	14.8	6.78	6.45	15.3	6.62	6.94	15.79	6.37	6.37	16.28	6.22	6.59	16.77	5.85	6.15	17.26	3.97	6.03	18	4.1	6.02	18.98	4.42	4.02	19.96	5.46	3.89	20.45	4.97	1.21	20.94	5.38	1.57	21.44	4.7	1.05	21.93	3.54	1.04	22.42	3.57	1.28	22.91	3.33	1.12	23.65	2.25	0	24.63	1.24	0.15	25.62	0.08	0.4	26.6	0.05	0.37	Bisakah kamu memberi label pada fase-fase dalam tabel data eksperimen waist trainer, menunjukkan mana yang merupakan fase peningkatan, stabil, dan penurunan?																																																																																											
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14		Berdasarkan gambar outdoor fitness double wheel dengan parameter yang diberikan, dapatkah kamu menjelaskan peran dari jari-jari (r), kecepatan sudut (ω), dan percepatan sentripetal (a) dalam fungsionalitas roda tersebut? Jelaskan juga bagaimana ketiga variabel ini saling berinteraksi dalam konteks gerak rotasi.																																																																																																																																																																																																																												
15		Kamu sedang mengamati outdoor fitness double wheel. Roda tersebut berputar dalam lintasan melingkar. Jarak dari pusat roda ke tepi bilah adalah jari-jari (r) 0,43 m dan kecepatan rotasinya adalah kecepatan sudut (ω). Dengan mempertimbangkan istilah-istilah ini, bisakah kamu menjelaskan bagaimana percepatan sentripetal (a) dari bilah dipengaruhi oleh jari-jari dan kecepatan sudut?																																																																																																																																																																																																																												
16	<table><tr><th>Time (s)</th><th>Acceleration (m/s^2)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>0.33</td><td>0.01</td><td>0</td><td>Increase</td></tr><tr><td>0.82</td><td>0</td><td>0</td><td>Increase</td></tr><tr><td>1.31</td><td>0.01</td><td>0.32</td><td>Increase</td></tr><tr><td>2.3</td><td>4.05</td><td>1.94</td><td>Increase</td></tr><tr><td>3.28</td><td>3.45</td><td>2.48</td><td>Increase</td></tr><tr><td>4.26</td><td>4.63</td><td>4.29</td><td>Increase</td></tr><tr><td>5.24</td><td>7.63</td><td>5.69</td><td>Increase</td></tr><tr><td>6.23</td><td>7.48</td><td>4.67</td><td>Increase</td></tr><tr><td>7.21</td><td>18.27</td><td>6.29</td><td></td></tr><tr><td>8.19</td><td>15.82</td><td>6.22</td><td></td></tr><tr><td>8.69</td><td>15.3</td><td>6.18</td><td></td></tr><tr><td>9.18</td><td>13.47</td><td>5.85</td><td></td></tr><tr><td>10.16</td><td>27.12</td><td>7.74</td><td>Steady</td></tr><tr><td>11.38</td><td>27.01</td><td>7.36</td><td>Steady</td></tr><tr><td>12.37</td><td>22.91</td><td>6.98</td><td>Steady</td></tr><tr><td>13.35</td><td>18.61</td><td>6.63</td><td>Steady</td></tr><tr><td>14.33</td><td>16.18</td><td>6.42</td><td>Steady</td></tr><tr><td>14.83</td><td>16.83</td><td>6.35</td><td>Steady</td></tr><tr><td>15.32</td><td>22.28</td><td>6.2</td><td>Steady</td></tr><tr><td>15.81</td><td>11.21</td><td>6.08</td><td>Steady</td></tr><tr><td>16.3</td><td>19.56</td><td>5.99</td><td>Steady</td></tr><tr><td>16.79</td><td>10.87</td><td>5.71</td><td>Steady</td></tr><tr><td>17.28</td><td>14.8</td><td>5.78</td><td>Steady</td></tr><tr><td>17.77</td><td>11.21</td><td>5.84</td><td></td></tr><tr><td>18.27</td><td>11.86</td><td>5.5</td><td></td></tr><tr><td>18.76</td><td>3.94</td><td>5.13</td><td></td></tr><tr><td>19.25</td><td>13.97</td><td>5.04</td><td></td></tr><tr><td>19.98</td><td>9.8</td><td>4.58</td><td></td></tr><tr><td>20.97</td><td>13.31</td><td>4.57</td><td></td></tr><tr><td>21.95</td><td>7.6</td><td>4.37</td><td></td></tr><tr><td>22.44</td><td>9.99</td><td>4.19</td><td></td></tr><tr><td>22.93</td><td>3.93</td><td>3.71</td><td></td></tr><tr><td>23.42</td><td>5.64</td><td>3.9</td><td></td></tr><tr><td>23.91</td><td>5.52</td><td>3.92</td><td></td></tr><tr><td>24.41</td><td>9.45</td><td>3.37</td><td></td></tr><tr><td>24.9</td><td>4.34</td><td>3.21</td><td></td></tr><tr><td>25.39</td><td>2.05</td><td>3.52</td><td></td></tr><tr><td>25.88</td><td>6.53</td><td>3.36</td><td></td></tr><tr><td>26.37</td><td>5.42</td><td>2.77</td><td></td></tr><tr><td>26.86</td><td>1.78</td><td>2.57</td><td></td></tr><tr><td>27.36</td><td>6.53</td><td>2.88</td><td></td></tr><tr><td>27.85</td><td>6.34</td><td>3.01</td><td></td></tr><tr><td>28.34</td><td>1.62</td><td>2.52</td><td></td></tr><tr><td>28.83</td><td>1.7</td><td>1.92</td><td></td></tr><tr><td>29.56</td><td>2.27</td><td>1.85</td><td></td></tr><tr><td>30.55</td><td>3.77</td><td>2.38</td><td>Decrease</td></tr><tr><td>31.53</td><td>0.97</td><td>1.35</td><td>Decrease</td></tr><tr><td>32.27</td><td>0.15</td><td>0.77</td><td>Decrease</td></tr><tr><td>32.76</td><td>0.27</td><td>0.37</td><td>Decrease</td></tr><tr><td>33.25</td><td>0.89</td><td>0.09</td><td>Decrease</td></tr><tr><td>33.75</td><td>0.62</td><td>0.03</td><td>Decrease</td></tr><tr><td>34.24</td><td>0.64</td><td>0</td><td>Steady</td></tr><tr><td>34.73</td><td>0.32</td><td>0.04</td><td>Steady</td></tr><tr><td>35.22</td><td>0.3</td><td>0.01</td><td>Steady</td></tr></table>	Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase	0.33	0.01	0	Increase	0.82	0	0	Increase	1.31	0.01	0.32	Increase	2.3	4.05	1.94	Increase	3.28	3.45	2.48	Increase	4.26	4.63	4.29	Increase	5.24	7.63	5.69	Increase	6.23	7.48	4.67	Increase	7.21	18.27	6.29		8.19	15.82	6.22		8.69	15.3	6.18		9.18	13.47	5.85		10.16	27.12	7.74	Steady	11.38	27.01	7.36	Steady	12.37	22.91	6.98	Steady	13.35	18.61	6.63	Steady	14.33	16.18	6.42	Steady	14.83	16.83	6.35	Steady	15.32	22.28	6.2	Steady	15.81	11.21	6.08	Steady	16.3	19.56	5.99	Steady	16.79	10.87	5.71	Steady	17.28	14.8	5.78	Steady	17.77	11.21	5.84		18.27	11.86	5.5		18.76	3.94	5.13		19.25	13.97	5.04		19.98	9.8	4.58		20.97	13.31	4.57		21.95	7.6	4.37		22.44	9.99	4.19		22.93	3.93	3.71		23.42	5.64	3.9		23.91	5.52	3.92		24.41	9.45	3.37		24.9	4.34	3.21		25.39	2.05	3.52		25.88	6.53	3.36		26.37	5.42	2.77		26.86	1.78	2.57		27.36	6.53	2.88		27.85	6.34	3.01		28.34	1.62	2.52		28.83	1.7	1.92		29.56	2.27	1.85		30.55	3.77	2.38	Decrease	31.53	0.97	1.35	Decrease	32.27	0.15	0.77	Decrease	32.76	0.27	0.37	Decrease	33.25	0.89	0.09	Decrease	33.75	0.62	0.03	Decrease	34.24	0.64	0	Steady	34.73	0.32	0.04	Steady	35.22	0.3	0.01	Steady	John sedang mengamati alat outdoor fitness double wheel. Ia melihat bahwa saat kecepatan sudut (ω) roda meningkat, percepatan sentripetal (a) juga ikut meningkat. Jika jari-jari roda adalah 0,43 meter, bagaimana hubungan antara ω dan a?
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17	<table><tr><th>Time (s)</th><th>Acceleration (m/s^2)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>0.33</td><td>0.01</td><td>0</td><td>Increase</td></tr><tr><td>0.82</td><td>0</td><td>0</td><td>Increase</td></tr><tr><td>1.31</td><td>0.01</td><td>0.32</td><td>Increase</td></tr><tr><td>2.3</td><td>4.05</td><td>1.94</td><td>Increase</td></tr><tr><td>3.28</td><td>3.45</td><td>2.48</td><td>Increase</td></tr><tr><td>4.26</td><td>4.63</td><td>4.29</td><td>Increase</td></tr><tr><td>5.24</td><td>7.63</td><td>5.69</td><td>Increase</td></tr><tr><td>6.23</td><td>7.48</td><td>4.67</td><td>Increase</td></tr><tr><td>7.21</td><td>18.27</td><td>6.29</td><td></td></tr><tr><td>8.19</td><td>15.82</td><td>6.22</td><td></td></tr><tr><td>8.69</td><td>15.3</td><td>6.18</td><td></td></tr><tr><td>9.18</td><td>13.47</td><td>5.85</td><td></td></tr><tr><td>10.16</td><td>27.12</td><td>7.74</td><td>Steady</td></tr><tr><td>11.38</td><td>27.01</td><td>7.36</td><td>Steady</td></tr><tr><td>12.37</td><td>22.91</td><td>6.98</td><td>Steady</td></tr><tr><td>13.35</td><td>18.61</td><td>6.63</td><td>Steady</td></tr><tr><td>14.33</td><td>16.18</td><td>6.42</td><td>Steady</td></tr><tr><td>14.83</td><td>16.83</td><td>6.35</td><td>Steady</td></tr><tr><td>15.32</td><td>22.28</td><td>6.2</td><td>Steady</td></tr><tr><td>15.81</td><td>11.21</td><td>6.08</td><td>Steady</td></tr><tr><td>16.3</td><td>19.56</td><td>5.99</td><td>Steady</td></tr><tr><td>16.79</td><td>10.87</td><td>5.71</td><td>Steady</td></tr><tr><td>17.28</td><td>14.8</td><td>5.78</td><td>Steady</td></tr><tr><td>17.77</td><td>11.21</td><td>5.84</td><td></td></tr><tr><td>18.27</td><td>11.86</td><td>5.5</td><td></td></tr><tr><td>18.76</td><td>3.94</td><td>5.13</td><td></td></tr><tr><td>19.25</td><td>13.97</td><td>5.04</td><td></td></tr><tr><td>19.98</td><td>9.8</td><td>4.58</td><td></td></tr><tr><td>20.97</td><td>13.31</td><td>4.57</td><td></td></tr><tr><td>21.95</td><td>7.6</td><td>4.37</td><td></td></tr><tr><td>22.44</td><td>9.99</td><td>4.19</td><td></td></tr><tr><td>22.93</td><td>3.93</td><td>3.71</td><td></td></tr><tr><td>23.42</td><td>5.64</td><td>3.9</td><td></td></tr><tr><td>23.91</td><td>5.52</td><td>3.92</td><td></td></tr><tr><td>24.41</td><td>9.45</td><td>3.37</td><td></td></tr><tr><td>24.9</td><td>4.34</td><td>3.21</td><td></td></tr><tr><td>25.39</td><td>2.05</td><td>3.52</td><td></td></tr><tr><td>25.88</td><td>6.53</td><td>3.36</td><td></td></tr><tr><td>26.37</td><td>5.42</td><td>2.77</td><td></td></tr><tr><td>26.86</td><td>1.78</td><td>2.57</td><td></td></tr><tr><td>27.36</td><td>6.53</td><td>2.88</td><td></td></tr><tr><td>27.85</td><td>6.34</td><td>3.01</td><td></td></tr><tr><td>28.34</td><td>1.62</td><td>2.52</td><td></td></tr><tr><td>28.83</td><td>1.7</td><td>1.92</td><td></td></tr><tr><td>29.56</td><td>2.27</td><td>1.85</td><td></td></tr><tr><td>30.55</td><td>3.77</td><td>2.38</td><td>Decrease</td></tr><tr><td>31.53</td><td>0.97</td><td>1.35</td><td>Decrease</td></tr><tr><td>32.27</td><td>0.15</td><td>0.77</td><td>Decrease</td></tr><tr><td>32.76</td><td>0.27</td><td>0.37</td><td>Decrease</td></tr><tr><td>33.25</td><td>0.89</td><td>0.09</td><td>Decrease</td></tr><tr><td>33.75</td><td>0.62</td><td>0.03</td><td>Decrease</td></tr><tr><td>34.24</td><td>0.64</td><td>0</td><td>Steady</td></tr><tr><td>34.73</td><td>0.32</td><td>0.04</td><td>Steady</td></tr><tr><td>35.22</td><td>0.3</td><td>0.01</td><td>Steady</td></tr></table>	Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase	0.33	0.01	0	Increase	0.82	0	0	Increase	1.31	0.01	0.32	Increase	2.3	4.05	1.94	Increase	3.28	3.45	2.48	Increase	4.26	4.63	4.29	Increase	5.24	7.63	5.69	Increase	6.23	7.48	4.67	Increase	7.21	18.27	6.29		8.19	15.82	6.22		8.69	15.3	6.18		9.18	13.47	5.85		10.16	27.12	7.74	Steady	11.38	27.01	7.36	Steady	12.37	22.91	6.98	Steady	13.35	18.61	6.63	Steady	14.33	16.18	6.42	Steady	14.83	16.83	6.35	Steady	15.32	22.28	6.2	Steady	15.81	11.21	6.08	Steady	16.3	19.56	5.99	Steady	16.79	10.87	5.71	Steady	17.28	14.8	5.78	Steady	17.77	11.21	5.84		18.27	11.86	5.5		18.76	3.94	5.13		19.25	13.97	5.04		19.98	9.8	4.58		20.97	13.31	4.57		21.95	7.6	4.37		22.44	9.99	4.19		22.93	3.93	3.71		23.42	5.64	3.9		23.91	5.52	3.92		24.41	9.45	3.37		24.9	4.34	3.21		25.39	2.05	3.52		25.88	6.53	3.36		26.37	5.42	2.77		26.86	1.78	2.57		27.36	6.53	2.88		27.85	6.34	3.01		28.34	1.62	2.52		28.83	1.7	1.92		29.56	2.27	1.85		30.55	3.77	2.38	Decrease	31.53	0.97	1.35	Decrease	32.27	0.15	0.77	Decrease	32.76	0.27	0.37	Decrease	33.25	0.89	0.09	Decrease	33.75	0.62	0.03	Decrease	34.24	0.64	0	Steady	34.73	0.32	0.04	Steady	35.22	0.3	0.01	Steady	Rina, seorang mahasiswa fisika, sedang menganalisis data dari eksperimen outdoor fitness double wheel dengan jari-jari bilah 0,43 m. Rina melihat bahwa saat alat outdoor fitness double wheel berputar lebih cepat, percepatan sentripetal juga meningkat. Namun, terkadang a tetap stabil meskipun ω berubah. Apa yang bisa memengaruhi hubungan ini?
Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase																																																																																																																																																																																																																											
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18		Selama periode stabil dari detik ke-10 hingga ke-17, kecepatan sudut (ω) dari outdoor fitness double wheel tetap stabil sekitar 6 hingga 7 rad/s. Jika jari-jari (r) dari roda adalah 0,43 m, bagaimana percepatan sentripetal (a) akan terpengaruh selama periode ini?																																																																																																																				
19		Selama periode stabil dari detik ke-30 hingga ke-35, outdoor fitness double wheel memiliki kecepatan sudut (ω) sekitar 1 rad/s dan jari-jari (r) sebesar 0,43 m. Apa yang akan terjadi pada percepatan sentripetal (a) jika kecepatan sudut (ω) menjadi dua kali lipat sementara jari-jari (r) tetap sama?																																																																																																																				
20		kamu sedang mengamati kipas listrik. Bilah-bilah kipas berputar dalam lintasan melingkar. Jarak dari pusat kipas ke ujung bilah adalah jari-jari (r), dan kecepatan rotasi bilah adalah kecepatan sudut (ω). Dengan mempertimbangkan istilah-istilah ini, dapatkan kamu menjelaskan bagaimana percepatan sentripetal (a) dari bilah kipas dipengaruhi oleh jari-jari dan kecepatan sudut?																																																																																																																				
21		Kamu sedang mengamati kipas listrik dengan jari-jari 0,17 m. Saat kipas berputar pada kecepatan tertingginya, ujung bilah bergerak dengan kecepatan sudut (ω). Dapatkan kamu menentukan percepatan sentripetal (a) dari ujung bilah tersebut? Harap nyatakan jawabanmu dalam bentuk ω .																																																																																																																				
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24		Diberikan roda sepeda dengan jari-jari (r) 0,2 meter dan pemahaman tentang kecepatan sudut (ω), dapatkah kamu menjelaskan bagaimana perubahan kecepatan sudut memengaruhi percepatan sentripetal (a) dari roda?																																																																																																																																													
25		Sebuah roda sepeda dengan jari-jari (r) 0,2 meter diamati selama periode waktu tertentu. Berdasarkan grafik yang diberikan dan analisis data eksperimen, dapatkah kamu menjelaskan bagaimana percepatan sentripetal (a) akan berperilaku selama periode yang berbeda saat kecepatan sudut (ω) baik stabil maupun berfluktuasi?																																																																																																																																													
26		Mengacu pada grafik dan observasi roda sepeda dengan jari-jari (r) 0,2 meter, dapatkah kamu merangkum hubungan antara kecepatan sudut (ω) dan percepatan sentripetal (a) selama periode stabil dan periode fluktuasi?																																																																																																																																													
27		Gambar menunjukkan mesin pencuci sayuran saat beroperasi. Bisakah kamu mengidentifikasi dan menjelaskan komponen dan konsep kunci dalam konteks fungsi alat tersebut, khususnya percepatan sentripetal (a), jari-jari (r), dan kecepatan sudut (ω)?																																																																																																																																													
28		Bisakah kamu menjelaskan bagaimana jari-jari (r) bilah mesin pencuci sayuran memengaruhi percepatan sentripetal (a) dan kecepatan sudut (ω)?																																																																																																																																													
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31		Dalam eksperimen mesin pencuci sayuran, bagaimana respons percepatan sentripetal (a) selama peningkatan awal kecepatan sudut (ω) dari 0s hingga 5s, periode stabil dari 10s hingga 20s saat ω tetap sekitar 14 rad/s, dan penurunan tajam dari 24s hingga 30s saat ω menurun mendekati nol?																																																																																																																																													
32		Diberikan data grafik dari mesin pencuci sayuran dengan jari-jari (r) 0,1 m, grafik menunjukkan bagaimana kecepatan sudut (ω) berubah seiring waktu dan dampaknya terhadap percepatan sentripetal (a). Grafik memperlihatkan bahwa dari 0 hingga sekitar 5 detik, ω meningkat cepat, yang menyebabkan peningkatan proporsional dalam a; dari 10 hingga 20 detik, ω stabil sekitar 14 rad/s, menghasilkan a yang stabil; dan setelah 25 detik, ω menurun tajam, menyebabkan a menurun. Bisakah kamu merangkum bagaimana fase-fase berbeda dari kecepatan sudut berkaitan dengan perubahan dalam percepatan sentripetal?																																																																																																																																													
33		Berdasarkan data yang diamati dari mesin pencuci sayuran dengan jari-jari 0,1 m, dapatkah kamu menjelaskan bagaimana perubahan percepatan sentripetal berkaitan dengan kecepatan sudut selama periode operasi yang berbeda?																																																																																																																																													
34		Diketahui mesin pencuci sayuran dengan jari-jari 0,1 meter dan perubahan kecepatan sudut yang diamati dari waktu ke waktu seperti yang ditunjukkan dalam grafik, bagaimana perilaku percepatan sentripetal selama periode peningkatan, stabil, dan penurunan sepanjang eksperimen 30 detik?																																																																																																																																													
35		Gambar menunjukkan mesin pencuci sayuran dengan jari-jari (r) 0,1 meter. Dalam konteks gerakan rotasinya, dapatkah kamu: Menjelaskan arti dari percepatan sentripetal (a) untuk komponen-komponen yang berputar di dalam alat tersebut?																																																																																																																																													

36		Gambar menunjukkan mesin pencuci sayuran dengan jari-jari (r) 0,1 meter. Dalam konteks gerakan rotasinya, dapatkah kamu: Menjelaskan bagaimana jari-jari (r) berhubungan dengan percepatan ke dalam yang dialami oleh bagian yang berputar?
37		Berdasarkan gambar rotating display plate, dapatkah kamu menjelaskan bagaimana jari-jari (r), percepatan sentripetal (a), dan kecepatan sudut (ω) ditampilkan secara visual dan bagaimana mereka saling terkait dalam mempertahankan gerak melingkar plate tersebut?
38		Diberikan gambar rotating display plate, dapatkah kamu mengidentifikasi dan menjelaskan fungsi dari percepatan sentripetal (a), jari-jari (r), dan kecepatan sudut (ω) berdasarkan anotasi visual? Harap jelaskan bagaimana masing-masing elemen berkontribusi terhadap gerakan melingkar yang diamati dalam sistem tersebut.
39		Berdasarkan grafik yang menunjukkan kecepatan sudut dan percepatan sentripetal dari rotating display plate, dapatkah kamu mengidentifikasi interval waktu spesifik di mana peningkatan paling signifikan dalam percepatan sentripetal terjadi?
40		Sebuah rotating display plate dengan jari-jari 0,14 m menunjukkan perubahan kecepatan sudut pada berbagai interval waktu. Antara detik ke-2 hingga ke-7, kecepatan sudut meningkat cepat, sedangkan antara detik ke-7 hingga ke-16 tetap stabil. Bagaimana perubahan spesifik dalam kecepatan sudut ini memengaruhi percepatan sentripetal dari piring tersebut selama periode ini?
41		Diberikan grafik dari rotating display plate dengan jari-jari 0,14 m, dapatkah kamu menjelaskan bagaimana percepatan sentripetal berubah dari waktu ke waktu dan bagaimana perubahan tersebut berkaitan dengan tren kecepatan sudut yang diamati dalam berbagai segmen waktu?
42		Berdasarkan grafik data rotating display plate dengan jari-jari 0,14 m, dapatkah kamu mengidentifikasi berbagai fase percepatan sentripetal dari waktu ke waktu—seperti fase stabil dan peningkatan selama interval 0 hingga 30 detik?

43	<table><tr><th>Time (s)</th><th>Acceleration (m/s^2)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>0.57</td><td>0.02</td><td>0</td><td></td></tr><tr><td>1.07</td><td>0.01</td><td>0</td><td></td></tr><tr><td>1.57</td><td>0</td><td>0</td><td></td></tr><tr><td>2.07</td><td>0.04</td><td>0.02</td><td></td></tr><tr><td>2.58</td><td>0.05</td><td>0</td><td></td></tr><tr><td>3.08</td><td>0</td><td>0</td><td>Increase</td></tr><tr><td>3.58</td><td>0</td><td>0.04</td><td>Increase</td></tr><tr><td>4.33</td><td>2.21</td><td>2.96</td><td>Increase</td></tr><tr><td>4.83</td><td>4.58</td><td>6.05</td><td>Increase</td></tr><tr><td>5.33</td><td>9.25</td><td>8.83</td><td>Increase</td></tr><tr><td>5.84</td><td>12.99</td><td>12.07</td><td>Increase</td></tr><tr><td>6.34</td><td>15.4</td><td>13.42</td><td>Increase</td></tr><tr><td>6.84</td><td>15.87</td><td>14.22</td><td>Increase</td></tr><tr><td>7.34</td><td>17.96</td><td>14.68</td><td>Increase</td></tr><tr><td>7.84</td><td>19.71</td><td>15.58</td><td>Increase</td></tr><tr><td>8.34</td><td>21.14</td><td>16.13</td><td>Increase</td></tr><tr><td>8.85</td><td>18.34</td><td>15.96</td><td>Steady</td></tr><tr><td>9.35</td><td>15.94</td><td>14.89</td><td>Steady</td></tr><tr><td>9.85</td><td>15.25</td><td>13.91</td><td>Steady</td></tr><tr><td>10.35</td><td>17.85</td><td>14.45</td><td>Steady</td></tr><tr><td>10.85</td><td>18.4</td><td>15.19</td><td>Steady</td></tr><tr><td>11.35</td><td>18.84</td><td>15.28</td><td>Steady</td></tr><tr><td>11.85</td><td>20.57</td><td>15.79</td><td>Steady</td></tr><tr><td>12.36</td><td>20.86</td><td>16.47</td><td>Steady</td></tr><tr><td>12.86</td><td>19.2</td><td>15.81</td><td>Steady</td></tr><tr><td>13.36</td><td>18.81</td><td>15.45</td><td>Steady</td></tr><tr><td>13.86</td><td>22.76</td><td>16.25</td><td>Steady</td></tr><tr><td>14.36</td><td>21.64</td><td>16.94</td><td>Steady</td></tr><tr><td>14.86</td><td>20.15</td><td>16.09</td><td>Steady</td></tr><tr><td>15.37</td><td>23.19</td><td>16.61</td><td>Steady</td></tr><tr><td>15.87</td><td>22.2</td><td>17.25</td><td>Steady</td></tr><tr><td>16.37</td><td>18.59</td><td>15.72</td><td>Steady</td></tr><tr><td>16.87</td><td>17.71</td><td>15.26</td><td>Decrease</td></tr><tr><td>17.37</td><td>16.09</td><td>14.63</td><td>Decrease</td></tr><tr><td>17.87</td><td>13.04</td><td>13.67</td><td>Decrease</td></tr><tr><td>18.37</td><td>9.27</td><td>11.68</td><td>Decrease</td></tr><tr><td>18.88</td><td>6.85</td><td>9.81</td><td>Decrease</td></tr><tr><td>19.38</td><td>5.53</td><td>8.8</td><td>Decrease</td></tr><tr><td>19.88</td><td>5.52</td><td>8.46</td><td>Decrease</td></tr><tr><td>20.38</td><td>5.67</td><td>8.42</td><td>Decrease</td></tr><tr><td>20.88</td><td>4.59</td><td>8.11</td><td>Decrease</td></tr><tr><td>21.38</td><td>4.03</td><td>7.47</td><td>Decrease</td></tr><tr><td>21.89</td><td>3.32</td><td>6.6</td><td>Decrease</td></tr></table>	Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase	0.57	0.02	0		1.07	0.01	0		1.57	0	0		2.07	0.04	0.02		2.58	0.05	0		3.08	0	0	Increase	3.58	0	0.04	Increase	4.33	2.21	2.96	Increase	4.83	4.58	6.05	Increase	5.33	9.25	8.83	Increase	5.84	12.99	12.07	Increase	6.34	15.4	13.42	Increase	6.84	15.87	14.22	Increase	7.34	17.96	14.68	Increase	7.84	19.71	15.58	Increase	8.34	21.14	16.13	Increase	8.85	18.34	15.96	Steady	9.35	15.94	14.89	Steady	9.85	15.25	13.91	Steady	10.35	17.85	14.45	Steady	10.85	18.4	15.19	Steady	11.35	18.84	15.28	Steady	11.85	20.57	15.79	Steady	12.36	20.86	16.47	Steady	12.86	19.2	15.81	Steady	13.36	18.81	15.45	Steady	13.86	22.76	16.25	Steady	14.36	21.64	16.94	Steady	14.86	20.15	16.09	Steady	15.37	23.19	16.61	Steady	15.87	22.2	17.25	Steady	16.37	18.59	15.72	Steady	16.87	17.71	15.26	Decrease	17.37	16.09	14.63	Decrease	17.87	13.04	13.67	Decrease	18.37	9.27	11.68	Decrease	18.88	6.85	9.81	Decrease	19.38	5.53	8.8	Decrease	19.88	5.52	8.46	Decrease	20.38	5.67	8.42	Decrease	20.88	4.59	8.11	Decrease	21.38	4.03	7.47	Decrease	21.89	3.32	6.6	Decrease	Rotating display plate dengan jari-jari (r) 0,14 m, serta perubahan kecepatan sudut (ω) dan percepatan sentripetal (a) dari waktu ke waktu, bisakah kamu menjelaskan bagaimana fase stabil dan puncak dari sistem tersebut menggambarkan hubungan antara kecepatan sudut dan percepatan sentripetal?
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2.58	0.05	0																																																																																																																																																																																
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11.35	18.84	15.28	Steady																																																																																																																																																																															
11.85	20.57	15.79	Steady																																																																																																																																																																															
12.36	20.86	16.47	Steady																																																																																																																																																																															
12.86	19.2	15.81	Steady																																																																																																																																																																															
13.36	18.81	15.45	Steady																																																																																																																																																																															
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21.89	3.32	6.6	Decrease																																																																																																																																																																															
44	<table><tr><th>Time (s)</th><th>Acceleration (m/s^2)</th><th>Angular Velocity (rad/s)</th><th>Phase</th></tr><tr><td>0.57</td><td>0.02</td><td>0</td><td></td></tr><tr><td>1.07</td><td>0.01</td><td>0</td><td></td></tr><tr><td>1.57</td><td>0</td><td>0</td><td></td></tr><tr><td>2.07</td><td>0.04</td><td>0.02</td><td></td></tr><tr><td>2.58</td><td>0.05</td><td>0</td><td></td></tr><tr><td>3.08</td><td>0</td><td>0</td><td>Increase</td></tr><tr><td>3.58</td><td>0</td><td>0.04</td><td>Increase</td></tr><tr><td>4.33</td><td>2.21</td><td>2.96</td><td>Increase</td></tr><tr><td>4.83</td><td>4.58</td><td>6.05</td><td>Increase</td></tr><tr><td>5.33</td><td>9.25</td><td>8.83</td><td>Increase</td></tr><tr><td>5.84</td><td>12.99</td><td>12.07</td><td>Increase</td></tr><tr><td>6.34</td><td>15.4</td><td>13.42</td><td>Increase</td></tr><tr><td>6.84</td><td>15.87</td><td>14.22</td><td>Increase</td></tr><tr><td>7.34</td><td>17.96</td><td>14.68</td><td>Increase</td></tr><tr><td>7.84</td><td>19.71</td><td>15.58</td><td>Increase</td></tr><tr><td>8.34</td><td>21.14</td><td>16.13</td><td>Increase</td></tr><tr><td>8.85</td><td>18.34</td><td>15.96</td><td>Steady</td></tr><tr><td>9.35</td><td>15.94</td><td>14.89</td><td>Steady</td></tr><tr><td>9.85</td><td>15.25</td><td>13.91</td><td>Steady</td></tr><tr><td>10.35</td><td>17.85</td><td>14.45</td><td>Steady</td></tr><tr><td>10.85</td><td>18.4</td><td>15.19</td><td>Steady</td></tr><tr><td>11.35</td><td>18.84</td><td>15.28</td><td>Steady</td></tr><tr><td>11.85</td><td>20.57</td><td>15.79</td><td>Steady</td></tr><tr><td>12.36</td><td>20.86</td><td>16.47</td><td>Steady</td></tr><tr><td>12.86</td><td>19.2</td><td>15.81</td><td>Steady</td></tr><tr><td>13.36</td><td>18.81</td><td>15.45</td><td>Steady</td></tr><tr><td>13.86</td><td>22.76</td><td>16.25</td><td>Steady</td></tr><tr><td>14.36</td><td>21.64</td><td>16.94</td><td>Steady</td></tr><tr><td>14.86</td><td>20.15</td><td>16.09</td><td>Steady</td></tr><tr><td>15.37</td><td>23.19</td><td>16.61</td><td>Steady</td></tr><tr><td>15.87</td><td>22.2</td><td>17.25</td><td>Steady</td></tr><tr><td>16.37</td><td>18.59</td><td>15.72</td><td>Steady</td></tr><tr><td>16.87</td><td>17.71</td><td>15.26</td><td>Decrease</td></tr><tr><td>17.37</td><td>16.09</td><td>14.63</td><td>Decrease</td></tr><tr><td>17.87</td><td>13.04</td><td>13.67</td><td>Decrease</td></tr><tr><td>18.37</td><td>9.27</td><td>11.68</td><td>Decrease</td></tr><tr><td>18.88</td><td>6.85</td><td>9.81</td><td>Decrease</td></tr><tr><td>19.38</td><td>5.53</td><td>8.8</td><td>Decrease</td></tr><tr><td>19.88</td><td>5.52</td><td>8.46</td><td>Decrease</td></tr><tr><td>20.38</td><td>5.67</td><td>8.42</td><td>Decrease</td></tr><tr><td>20.88</td><td>4.59</td><td>8.11</td><td>Decrease</td></tr><tr><td>21.38</td><td>4.03</td><td>7.47</td><td>Decrease</td></tr><tr><td>21.89</td><td>3.32</td><td>6.6</td><td>Decrease</td></tr></table>	Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase	0.57	0.02	0		1.07	0.01	0		1.57	0	0		2.07	0.04	0.02		2.58	0.05	0		3.08	0	0	Increase	3.58	0	0.04	Increase	4.33	2.21	2.96	Increase	4.83	4.58	6.05	Increase	5.33	9.25	8.83	Increase	5.84	12.99	12.07	Increase	6.34	15.4	13.42	Increase	6.84	15.87	14.22	Increase	7.34	17.96	14.68	Increase	7.84	19.71	15.58	Increase	8.34	21.14	16.13	Increase	8.85	18.34	15.96	Steady	9.35	15.94	14.89	Steady	9.85	15.25	13.91	Steady	10.35	17.85	14.45	Steady	10.85	18.4	15.19	Steady	11.35	18.84	15.28	Steady	11.85	20.57	15.79	Steady	12.36	20.86	16.47	Steady	12.86	19.2	15.81	Steady	13.36	18.81	15.45	Steady	13.86	22.76	16.25	Steady	14.36	21.64	16.94	Steady	14.86	20.15	16.09	Steady	15.37	23.19	16.61	Steady	15.87	22.2	17.25	Steady	16.37	18.59	15.72	Steady	16.87	17.71	15.26	Decrease	17.37	16.09	14.63	Decrease	17.87	13.04	13.67	Decrease	18.37	9.27	11.68	Decrease	18.88	6.85	9.81	Decrease	19.38	5.53	8.8	Decrease	19.88	5.52	8.46	Decrease	20.38	5.67	8.42	Decrease	20.88	4.59	8.11	Decrease	21.38	4.03	7.47	Decrease	21.89	3.32	6.6	Decrease	Berdasarkan data yang diamati dari rotating display plate dengan jari-jari tetap, jelaskan bagaimana percepatan sentripetal (a) berubah saat kecepatan sudut (ω) meningkat.
Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase																																																																																																																																																																															
0.57	0.02	0																																																																																																																																																																																
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2.07	0.04	0.02																																																																																																																																																																																
2.58	0.05	0																																																																																																																																																																																
3.08	0	0	Increase																																																																																																																																																																															
3.58	0	0.04	Increase																																																																																																																																																																															
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10.85	18.4	15.19	Steady																																																																																																																																																																															
11.35	18.84	15.28	Steady																																																																																																																																																																															
11.85	20.57	15.79	Steady																																																																																																																																																																															
12.36	20.86	16.47	Steady																																																																																																																																																																															
12.86	19.2	15.81	Steady																																																																																																																																																																															
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14.36	21.64	16.94	Steady																																																																																																																																																																															
14.86	20.15	16.09	Steady																																																																																																																																																																															
15.37	23.19	16.61	Steady																																																																																																																																																																															
15.87	22.2	17.25	Steady																																																																																																																																																																															
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16.87	17.71	15.26	Decrease																																																																																																																																																																															
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21.38	4.03	7.47	Decrease																																																																																																																																																																															
21.89	3.32	6.6	Decrease																																																																																																																																																																															
45		Kamu sedang mengamati kipas listrik dengan jari-jari 0,13 m. Saat kipas berputar pada kecepatan tertinggi, ujung bilahnya bergerak dengan kecepatan sudut (ω). Dapatkah kamu menentukan percepatan sentripetal (a) dari ujung bilah tersebut? Harap nyatakan jawabanmu dalam bentuk ω.																																																																																																																																																																																
46		Gambar menunjukkan sebuah sepeda dengan jari-jari (r) sebesar 0,2 m. Pada awal eksperimen, kecepatan sudut (ω) stabil pada 0,08 rad/s, menghasilkan percepatan sentripetal (a) sebesar 0,12 m/s². Pada akhir eksperimen, roda mencapai kecepatan sudut stabil sebesar 7,44 rad/s, dengan percepatan sentripetal 12,97 m/s². Ringkaslah hubungan antara jari-jari, kecepatan sudut, dan percepatan sentripetal seperti yang ditunjukkan dalam eksperimen ini.																																																																																																																																																																																

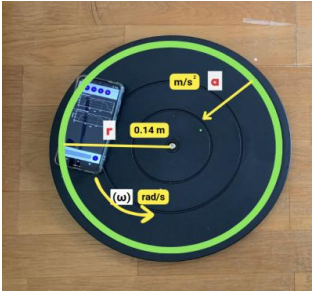
47



Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)
0.34	0.12	0.08
0.83	0.6	0.59
1.32	1.08	1.02
2.31	1.11	1.57
3.29	2.57	2.52
4.27	1.1	1.96
5.26	1.24	4.45
6.24	2.82	3.64
7.22	7.85	5.13
8.21	12.18	6.89
9.19	13.03	7.94
9.68	12.85	8.05
10.17	8.15	7.77
10.9	12.04	7.55
11.89	8.6	7.41
12.87	7.62	7.26
13.36	11.66	6.79
13.85	8.54	7.05
14.35	8.38	6.54
14.84	9.68	7.27
15.33	6.8	6.9
15.82	11.21	7.23
16.31	5.11	6.78
16.8	12.21	6.93
17.29	5.91	6.56
17.79	11.75	6.73
18.28	6.17	6.87
18.77	12.97	7.44
19.26	9.51	7.43
19.75	9.48	7.09
20.48	10.75	6.81
21.47	10.96	6.56
22.45	10.29	6.31
22.94	4.99	6.66
23.43	9.4	6.07
23.93	5.04	6.47
24.42	8.78	5.84
24.91	4.48	6.23
25.4	8.45	5.66
25.89	3.38	5.89
26.38	8.19	5.63
26.87	2.3	5.4
27.37	7.42	5.69
27.86	3.07	4.96
28.35	5.51	5.51
28.84	5.86	4.95

Berdasarkan data eksperimen dan gambar roda sepeda, jelaskan bagaimana perubahan percepatan sentripetal (a) dan kecepatan sudut (ω) terpengaruh ketika jari-jari roda (r) ditingkatkan?

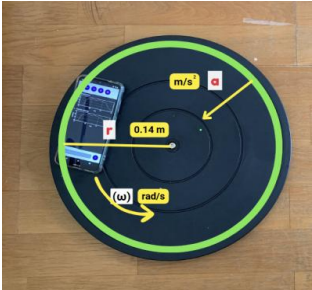
48



Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase
0.57	0.02	0	
1.07	0.01	0	
1.57	0	0	
2.07	0.04	0.02	
2.58	0.05	0	
3.08	0	0	Increase
3.58	0	0.04	Increase
4.33	2.21	2.96	Increase
4.83	4.58	6.05	Increase
5.33	9.25	8.83	Increase
5.84	12.99	12.07	Increase
6.34	15.4	13.42	Increase
6.84	15.87	14.22	Increase
7.34	17.96	14.68	Increase
7.84	19.71	15.58	Increase
8.34	21.14	16.13	Increase
8.85	18.34	15.96	Steady
9.35	15.94	14.89	Steady
9.85	15.25	13.91	Steady
10.35	17.85	14.45	Steady
10.85	18.4	15.19	Steady
11.35	18.84	15.28	Steady
11.85	20.57	15.79	Steady
12.36	20.86	16.47	Steady
12.86	19.2	15.81	Steady
13.36	18.81	15.45	Steady
13.86	22.76	16.25	Steady
14.36	21.64	16.94	Steady
14.86	20.15	16.09	Steady
15.37	23.19	16.61	Steady
15.87	22.2	17.25	Steady
16.37	18.59	15.72	Steady
16.87	17.71	15.26	Decrease
17.37	16.09	14.63	Decrease
17.87	13.04	13.67	Decrease
18.37	9.27	11.68	Decrease
18.88	6.85	9.81	Decrease
19.38	5.53	8.8	Decrease
19.88	5.52	8.46	Decrease
20.38	5.67	8.42	Decrease
20.88	4.59	8.11	Decrease
21.38	4.03	7.47	Decrease
21.89	3.32	6.6	Decrease

Berdasarkan perubahan kecepatan sudut dan gambar anotasi dari piring pajangan berputar, dapatkah kamu menjelaskan bagaimana arah dan besar percepatan sentripetal (a) terpengaruh selama berbagai fase gerakan?

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Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase
0.57	0.02	0	
1.07	0.01	0	
1.57	0	0	
2.07	0.04	0.02	
2.58	0.05	0	
3.08	0	0	Increase
3.58	0	0.04	Increase
4.33	2.21	2.96	Increase
4.83	4.58	6.05	Increase
5.33	9.25	8.83	Increase
5.84	12.99	12.07	Increase
6.34	15.4	13.42	Increase
6.84	15.87	14.22	Increase
7.34	17.96	14.68	Increase
7.84	19.71	15.58	Increase
8.34	21.14	16.13	Increase
8.85	18.34	15.96	Steady
9.35	15.94	14.89	Steady
9.85	15.25	13.91	Steady
10.35	17.85	14.45	Steady
10.85	18.4	15.19	Steady
11.35	18.84	15.28	Steady
11.85	20.57	15.79	Steady
12.36	20.86	16.47	Steady
12.86	19.2	15.81	Steady
13.36	18.81	15.45	Steady
13.86	22.76	16.25	Steady
14.36	21.64	16.94	Steady
14.86	20.15	16.09	Steady
15.37	23.19	16.61	Steady
15.87	22.2	17.25	Steady
16.37	18.59	15.72	Steady
16.87	17.71	15.26	Decrease
17.37	16.09	14.63	Decrease
17.87	13.04	13.67	Decrease
18.37	9.27	11.68	Decrease
18.88	6.85	9.81	Decrease
19.38	5.53	8.8	Decrease
19.88	5.52	8.46	Decrease
20.38	5.67	8.42	Decrease
20.88	4.59	8.11	Decrease
21.38	4.03	7.47	Decrease
21.89	3.32	6.6	Decrease

Bagaimana perubahan kecepatan sudut yang diamati antara detik ke-3 hingga ke-30 memengaruhi arah dan besar percepatan sentripetal pada piring pajangan berputar seperti yang digambarkan dalam gambar?

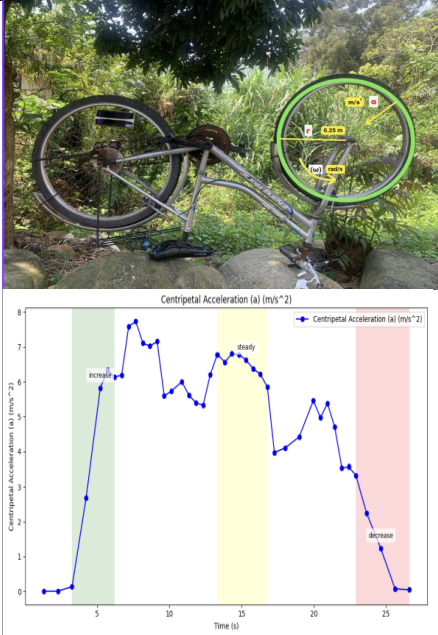
50



Time (s)	Acceleration (m/s^2)	Angular Velocity
0.33	0	0
1.31	0	0
2.29	0.04	0
3.28	0	0
4.26	2.14	6.07
5.24	11.99	10.28
6.23	13.75	11.22
6.72	14.38	11.57
7.21	15.33	12.25
7.7	16.34	12.77
8.19	15.15	12.1
8.68	15.14	12.03
9.17	15.09	12.01
9.67	16.38	12.69
10.16	16.95	13.09
10.89	18.32	14.55
11.38	17.82	13.89
11.87	17.24	13.07
12.36	16.56	12.86
12.86	16.7	12.98
13.35	16.47	12.48
13.84	15.76	12.41
14.33	15.68	12.17
14.82	15.6	12.71
15.31	17.42	13.56
15.81	17.61	13.43
16.3	17.41	13.14
16.79	16.71	13.16
17.28	17.74	13.96
18.01	18.33	12.67
19	15.75	12.97
19.98	17.63	12.68
20.47	16.52	11.85
20.96	15.09	11.54
21.45	14.36	11.85
21.94	14.92	11.8
22.44	14.74	11.61
22.93	14.59	11.5
23.67	12.51	10.42
24.65	10	8.65
25.64	7.25	6.4
26.62	5.32	3.82
27.84	4.05	0.02
28.83	0.93	0
29.81	0.42	0.17
30.3	0.17	0

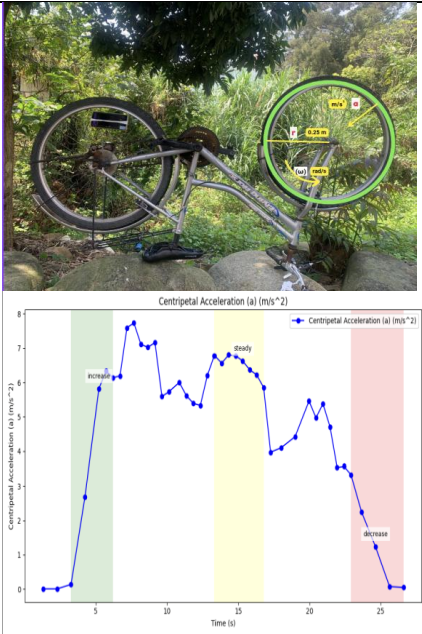
Berdasarkan gambar mesin pencuci sayuran yang menunjukkan jari-jari (r), kecepatan sudut (ω), dan percepatan sentripetal (a), bagaimana ketiga hal itu saling berhubungan saat mesin mulai berputar hingga berputar stabil?

51



Gambar menunjukkan roda sepeda dengan jari-jari (r) 0,25 m. Menurut grafik, kecepatan sudut (ω) tetap stabil selama periode stabil 5s–7s, 11s–13s, dan 16s–18s, yang menghasilkan percepatan sentripetal (a) yang konsisten. Jika kecepatan sudut (ω) mulai menurun tajam seperti yang diamati dari detik ke-18 hingga ke-25, bagaimana hal ini akan memengaruhi percepatan sentripetal (a) dari roda sepeda, dan mengapa?

52



Diberikan gambar roda sepeda berputar dan interpretasi grafik yang sesuai, jika roda (dengan jari-jari 0,25 m) mempertahankan kecepatan sudut (ω) stabil sebesar 6,5 rad/s dari detik ke-10 hingga ke-11, bagaimana kamu akan menggambarkan perilaku percepatan sentripetal (a) selama periode 12s hingga 14s, dengan asumsi ω tetap tidak berubah?

53

Berdasarkan gambar dan data eksperimen dari mesin pencuci sayuran, bagaimana elemen-elemen yang diberi label—jari-jari ($r = 0,1 \text{ m}$), kecepatan sudut (ω), dan percepatan sentripetal (a)—serta perubahan ω yang diamati dari 0s hingga 30s, menunjukkan hubungan antara ω dan a selama fase percepatan, stabil, dan perlambatan rotasi alat?

54

Bagaimana fitur yang dianotasi dalam gambar mesin pencuci sayuran dan tren data berbasis waktu secara bersama-sama menunjukkan hubungan antara kecepatan sudut (ω) dan percepatan sentripetal (a) selama berbagai fase gerakan?

55

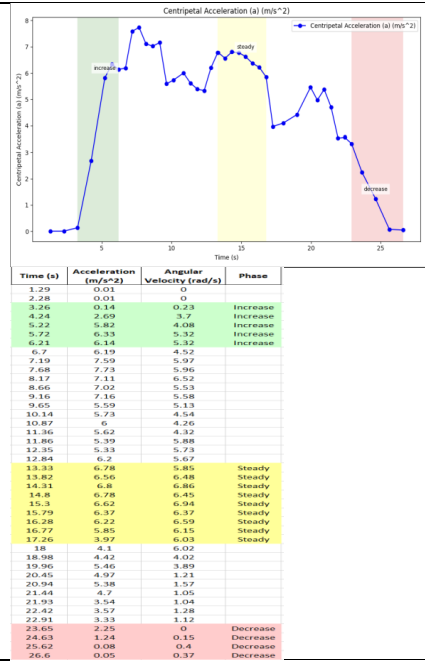
Bagaimana gambar anotasi kipas—yang menyoroti jari-jari ($r = 0,17 \text{ m}$), kecepatan sudut (ω), dan percepatan sentripetal (a)—bersama dengan data eksperimen dari 0s hingga 25s, menunjukkan perubahan dalam percepatan sentripetal selama periode kecepatan sudut yang meningkat, stabil, dan menurun?

56

Time (s)	Acceleration (m/s²)	Angular Velocity (rad/s)	Phase
1.29	0.01	0	
2.28	0.01	0	
3.26	0.14	0.23	Increase
4.24	2.69	3.7	Increase
5.22	5.82	4.08	Increase
5.72	6.33	5.32	Increase
6.21	6.14	5.32	Increase
6.7	6.19	4.52	
7.19	7.59	5.97	
7.68	7.73	5.96	
8.17	7.11	6.52	
8.66	7.02	5.53	
9.16	7.16	5.98	
9.65	5.59	5.13	
10.14	5.73	4.54	
10.87	6	4.26	
11.36	5.62	4.32	
11.86	5.39	5.88	
12.35	5.33	5.73	
12.84	6.2	5.67	
13.33	6.78	5.85	Steady
13.82	6.56	6.48	Steady
14.31	6.8	6.86	Steady
14.8	6.78	6.45	Steady
15.3	6.62	6.94	Steady
15.79	6.37	6.37	Steady
16.28	6.22	6.59	Steady
16.77	5.85	6.15	Steady
17.26	3.97	6.03	Steady
18	4.1	6.02	
18.98	4.42	4.02	
19.96	5.46	3.89	
20.45	4.97	1.23	
20.94	5.38	1.57	
21.44	4.7	1.05	
21.93	3.54	1.04	
22.42	3.57	1.28	
22.91	3.33	1.12	
23.65	2.25	0	Decrease
24.63	1.24	0.15	Decrease
25.62	0.08	0.4	Decrease
26.6	0.05	0.37	Decrease

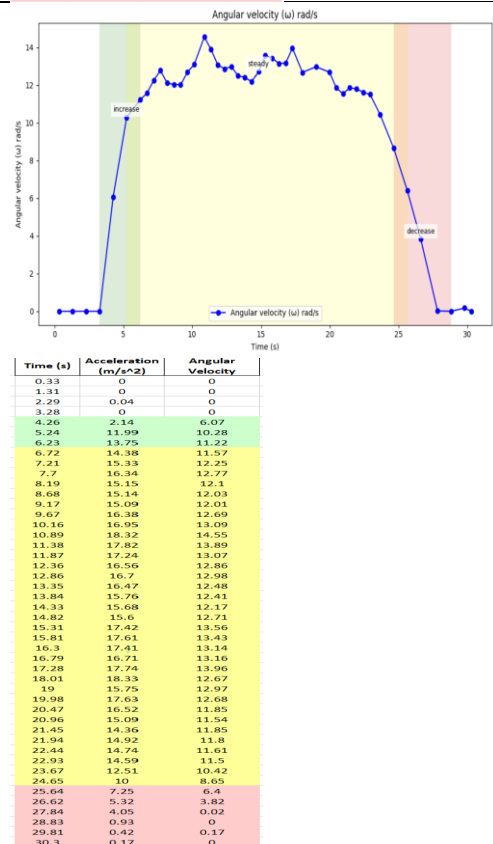
Mengacu pada tabel dan grafik, selama interval waktu manakah roda sepeda dengan jari-jari (r) tampak bergerak dalam lintasan melingkar yang stabil?

57



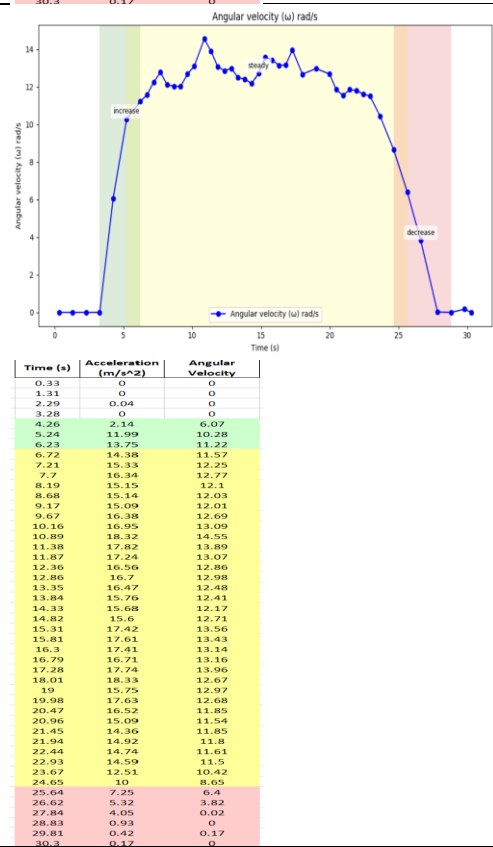
Berdasarkan tabel data dan grafik, selama periode waktu manakah kecepatan sudut (ω) dari roda sepeda tampak stabil, dan bagaimana stabilitas ini memengaruhi percepatan sentripetal (a) dan jari-jari (r)?

58



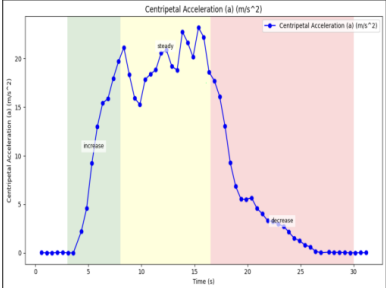
Bagaimana periode awal diam, rotasi stabil, dan perlambatan akhir dari mesin pencuci sayuran—berdasarkan grafik dan nilai dalam tabel kecepatan sudut (ω) dan percepatan sentripetal (a)—dengan jelas mendukung hubungan kuadrat antara ω dan a dalam gerak melingkar?

59



Bagaimana perubahan kecepatan sudut (ω) dan percepatan sentripetal (a) selama fase waktu yang berbeda dari eksperimen mesin pencuci sayuran mencerminkan hubungan langsung di antara keduanya, khususnya selama tahap peningkatan, stabil, dan penurunan gerakan?

Time (s)	Acceleration (m/s^2)	Angular Velocity (rad/s)	Phase
0.57	0.02	0	
1.07	0.01	0	
1.57	0	0	
2.07	0.04	0.02	
2.58	0.05	0	
3.08	0	0	Increase
3.58	0	0.04	Increase
4.33	2.21	2.96	Increase
4.83	4.58	6.05	Increase
5.33	9.25	8.83	Increase
5.84	12.99	12.07	Increase
6.34	15.4	13.42	Increase
6.84	15.87	14.22	Increase
7.34	17.96	14.68	Increase
7.84	19.71	15.58	Increase
8.34	21.14	16.13	Increase
8.85	18.34	15.96	Steady
9.35	15.94	14.89	Steady
9.85	15.25	13.91	Steady
10.35	17.85	14.45	Steady
10.85	18.4	15.19	Steady
11.35	18.84	15.28	Steady
11.85	20.57	15.79	Steady
12.36	20.86	16.47	Steady
12.86	19.2	15.81	Steady
13.36	18.81	15.45	Steady
13.86	22.76	16.25	Steady
14.36	21.64	16.94	Steady
14.86	20.15	16.09	Steady
15.37	23.19	16.61	Steady
15.87	22.2	17.25	Steady
16.37	18.59	15.72	Steady
16.87	17.71	15.26	Decrease
17.37	16.09	14.63	Decrease
17.87	13.04	13.67	Decrease
18.37	9.27	11.68	Decrease
18.88	6.85	9.81	Decrease
19.38	5.53	8.8	Decrease
19.88	5.52	8.46	Decrease
20.38	5.67	8.42	Decrease
20.88	4.59	8.11	Decrease
21.38	4.03	7.47	Decrease
21.89	3.32	6.6	Decrease



Ini adalah data grafik dan tabel dari piring pajangan berputar. Selama fase mana kecepatan sudut diperkirakan tetap stabil?